

Two Days Workshop on Robotics and Artificial Intelligence

- **Duration :** 2-days
- **Date:** 5 - 6 June, 2023
- **Timing:** 10 am- 4 pm
- **Eligibility:** B.Tech.
- **Mode:** offline
- **Number of Seats:** 30

Google link and QR Code for Registration



<https://forms.gle/VVugJwZanzuf2xna7>



Contents

- ❑ *Introduction to robotics*
- ❑ *Kinematics and dynamics of robotic systems*
- ❑ *Learning based energy optimal motion planning of robotic systems (industrial manipulators, biped robots)*

Day	Session	
	Forenoon	Afternoon
Day 1	Introduction to robotics: Kinematics • Trajectory planning of industrial manipulator and biped robots. • DH parameters • Forward kinematics, and inverse kinematics • Joints position, velocity, and acceleration computation • Hands-on MATLAB session on above-discussed topics	Introduction to robotics: Dynamics • Dynamics of robotic systems by using recursive Newton-Euler method for computation of joints torque • Hands-on MATLAB session on topics discussed in this session
Day 2	Supervised Learning: Classification, Regression using multilayer feed forward neural networks, MATLAB implementation of machine learning techniques	Learning based energy optimal motion planning of robotic systems : Dataset creation, Supervised Learning, Real-time implementation of trained neural network for generation of optimal path



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